

RESEARCH ARTICLE

A Symmetry Strategy for Large Bandgap Quantum Anomalous Hall Insulators

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Received: 26 January 2026 | **Revised:** 14 May 2026 | **Accepted:** 9 June 2026

Keywords: bandgap | ferromagnetism | quantum anomalous Hall effect | symmetry | two-dimensional material

ABSTRACT

Quantum anomalous Hall insulators (QAHIs) are an ideal platform for exploring exotic quantum phenomena, yet typically depend on strong relativistic effects to achieve a substantial bandgap. Here, we propose a symmetry-lowering strategy for designing robust QAHIs without introducing heavy elements. Using the tetragonal space group as a prototypical system, we find that reducing symmetry from $P4/nmm$ to $Pmmn$ activates hybridization of distinct orbital sets near the Dirac cone, producing a large topological bandgap in correlated systems. First-principles calculations identify the $Pmmn$ VAs monolayer as a high-temperature QAH with a bandgap of 1018 meV, much larger than 857 meV of its $P4/nmm$ counterpart. This strategy also facilitates bandgap engineering in QAHIs via symmetry manipulation, such as strain and substrate applications.

1 | Introduction

Quantum anomalous Hall insulators (QAHIs) are two-dimensional (2D) topological insulators exhibiting spin-polarized chiral edge states and quantized Hall conductivities without a magnet, which are robust against disorders and impurities [1]. Since the fully spin-polarized electrons in edge states move in only one direction, electronic backscattering is prohibited in QAHIs, making them ideal for developing dissipationless devices with integrated information storage and logic operations. Moreover, QAHIs provide a platform for exploring exotic quantum phenomena [2], such as topological magnetoelectricity [3] and Majorana modes by combining with superconductors [1]. Despite extensive efforts devoted to materializing this effect [4–11], all experimentally reported QAHIs are subject to stringent conditions at cryogenic temperatures.

To raise the operational temperature of QAHIs, the materials need to meet two conditions: i) strong spin exchange coupling for maintaining a high ferromagnetic Curie temperature T_c ; ii) strong spin-orbit coupling (SOC) for opening a large topological bandgap E_g against thermal fluctuation. However, the two conditions conflict with each other in most compounds: the former requires 3d orbitals of relatively light transition metal atoms, while the latter usually exists in materials with heavy elements. This conflation challenges the material design of high-temperature QAHIs. Indeed, among numerous predicted QAHIs [12–44], only a few heavy element-containing materials combine an E_g greater than 250 meV and a ferromagnetic T_c above room temperature, including InFeSe [45], KTiSb(Bi) [35], $V_2WS(Se)_4$ [40], FeI [29], and TiTe monolayers [32]. Actually, T_c can be increased by introducing 3d transition metals, but concertedly widening E_g is notoriously difficult. Recently, a new strategy of enhancing E_g by electron correlation in compounds containing

elements with d orbital sets was proposed to relieve the use of heavy elements for high-temperature QAHI [31]; among the nine predicted candidates, the FeX ($X = \text{S, Se}$) family can exhibit SOC gaps exceeding 600 meV. However, this strategy is based on an assumption of a single orbital set near the Fermi level, which is not satisfied in most materials. Moreover, dynamical lattice distortion, such as the Jahn–Teller effect, lowers the lattice symmetry and releases the orbital multiplets, which may impair the correlation-induced enhancement of the SOC gap.

In this work, we propose a symmetry-lowering strategy to design high-temperature QAHI from nonheavy elements. Exemplified with a tetragonal lattice, we find that reducing the symmetry from $P4/nmm$ to $Pmmn$ activates hybridization of distinct d -orbital sets near the Dirac point, which then interplays with electron correlation to widen E_g . Guided by this theory, we identify three new candidates exhibiting T_c above room temperature and E_g above 110 meV by high-throughput first-principles calculations. Especially, the 2D $Pmmn$ VAs is a QAHI with a T_c of 850 K and an E_g of 1018 meV, much higher than the parameter set of 650 K and 181 meV for the 2D $P4/nmm$ TiTe containing heavy element. Moreover, the bandgap of the high-symmetry TiTe monolayer can be widened by symmetry-lowering manipulation, such as strain and substrate applications. This strategy opens a new way to manipulating other SOC-related matter states.

2 | Results and Discussion

2.1 | Theoretical Analysis

To introduce our theory, we first consider a Dirac cone composed of a single orbital set of spin-splitting d shell, which is gapped out by the first-order SOC effect with a nonzero polarization of orbital angular momentum. When the Dirac cone is half occupied, the Coulomb correlation within the orbital set enhances E_g via the orbital polarization, which has been discussed in ref. [31]. However, in most systems, the Dirac cone is involved with multiple orbital sets. We find that lowering the lattice symmetry can activate distinct orbital sets to hybridize within the same cone. This orbital hybridization can interplay with electron correlation to synergistically result in a greatly widened E_g in QAHI without heavy elements, as discussed below.

In 2D systems with strong exchange splitting, only two d -orbital sets, namely $E_1 = \{d_{xz}, d_{yz}\}$ and $E_2 = \{d_{xy}, d_{x^2-y^2}\}$, can activate the first-order SOC effect. Considering a Dirac cone contributed only by a spin-polarized E_2 set, SOC gaps out the cone by $\Delta E_0 = \lambda \hbar |\bar{L}_z|$, where $|\bar{L}_z|$ is the orbital polarization, and λ is the SOC strength. If the Dirac cone is half-filled, the mean-field electron correlation $\hat{H}_{C,2}$ within E_2 enhances the effective λ , as described in ref. [31]. The effective bandgap is given by,

$$\Delta E_{\text{eff}} = \Delta E_0 + \frac{U_{\text{eff}} |\bar{L}_z|}{2\hbar} \quad (1)$$

where U_{eff} is the effective on-site Coulomb interaction parameter. The schematic of this SOC gap enhancement is shown in Figure 1c.

When both the E_1 and E_2 are involved in hybridizations, the hybridized states read $|\alpha\rangle = A_1 |d_{yz}\rangle + A_2 |d_{xy}\rangle$ and $|\beta\rangle = B_1 |d_{xz}\rangle + B_2 |d_{x^2-y^2}\rangle$. The first-order SOC gaps out the Dirac cone by $\Delta E_0 = \lambda \hbar |\bar{L}_{\text{eff}}|$ with effective orbital polarization $|\bar{L}_{\text{eff}}|$. Meanwhile, the correlation term, $\hat{H}_C = \hat{H}_{C,1} + \hat{H}_{C,2} + \hat{H}_{C,12}$, now includes intra-set contribution $\hat{H}_{C,1} + \hat{H}_{C,2}$ and inter-set contribution $\hat{H}_{C,12}$ (Figure 1d). In the half-filling condition, we obtain the effective gap size as (see Note S1 for derivation).

$$\Delta E_{\text{eff}} = \sqrt{\left(\Delta E_0 + \frac{U_{\text{eff}} |\bar{L}_{\text{eff}}|}{2\hbar}\right)^2 + \frac{U_{\text{eff}}^2 \bar{\Delta}^2}{4}} \quad (2)$$

Apparently, the enhancement comprises two parts: the first part is just a modification on the known effective SOC Equation (1), and the second part represents a new enhancement effect resulting from an interplay of correlation and orbital hybridization through the hybridization parameter $\bar{\Delta}$ (see Note S1), which has been overlooked in all related references (see Refs. [31, 32, 35]). Even when $|\bar{L}_{\text{eff}}|$ is almost quenched by electronic hopping or dynamical lattice distortion, such enhancement effect by orbital hybridization can still survive to significantly widen E_g of QAHI.

To activate the hybridization effect, the system symmetry should be lowered. For example, in a tetragonal system with space group $P4/nmm$ (No. 129), four Dirac cones equally composed of the E_2 set are distributed in the Γ -X and Γ -Y high-symmetry lines without SOC, respectively (Figure 1a). Under the high C_{4z} symmetry, d_{xz} contribution of the cone in the Γ -Y line is transformed to d_{yz} of the cone in the Γ -X line, which separates d_{xz} and d_{yz} and, thus, suppresses the hybridization between orbital sets (Figure 1a). By lowering the symmetry to $Pmmn$, the two orbitals are no longer separated, thereby enabling the inter-set hybridization within Dirac cones (Figure 1b). As shown in Figure 1c,d, the symmetry lowering and orbital hybridization then activate the additional intra-set $\hat{H}_{C,1}$ and inter-set correlations $\hat{H}_{C,12}$ besides the original $\hat{H}_{C,2}$, thereby leading to a larger enhancement effect.

2.2 | Tight-Binding Model

To verify this theory, we construct a concrete 10-band tight-binding (TB) model for QAHI and numerically evaluate the bandgap enhancement. The Hamiltonian is constructed from five d orbitals on two sublattices, expressed as:

$$\hat{H}_{\text{TB}} = \hat{H}_4 + \hat{H}_2 + \hat{H}_C \quad (3)$$

where $\hat{H}_4$ is a 10-band TB model of $P4/nmm$ symmetry (Figure 2c, upper panel), including on-site energies, electronic hopping far to the next nearest neighbor, and first-order SOC; $\hat{H}_2$ introduces corrections on on-site energies and hopping terms when C_{4z} is broken, ensuring that $\hat{H}_4 + \hat{H}_2$ follows the $Pmmn$ symmetry (Figure 2c, lower panel). Details on the TB model are provided in Note S2. The band structures of $\hat{H}_4$ with and without SOC are depicted in Figure 2a, showing four Dirac cones in the Γ -X and Γ -Y lines (each have two). d_{xz} and d_{yz} separately contribute to the two pairs of cones in the Γ -Y and Γ -X lines, respectively, while E_2 is associated with all the four cones (Note S2.1 and Figure S1), as dictated by the C_{4z} symmetry. Including SOC opens a bandgap at the cones, yielding a QAHI with a Chern number

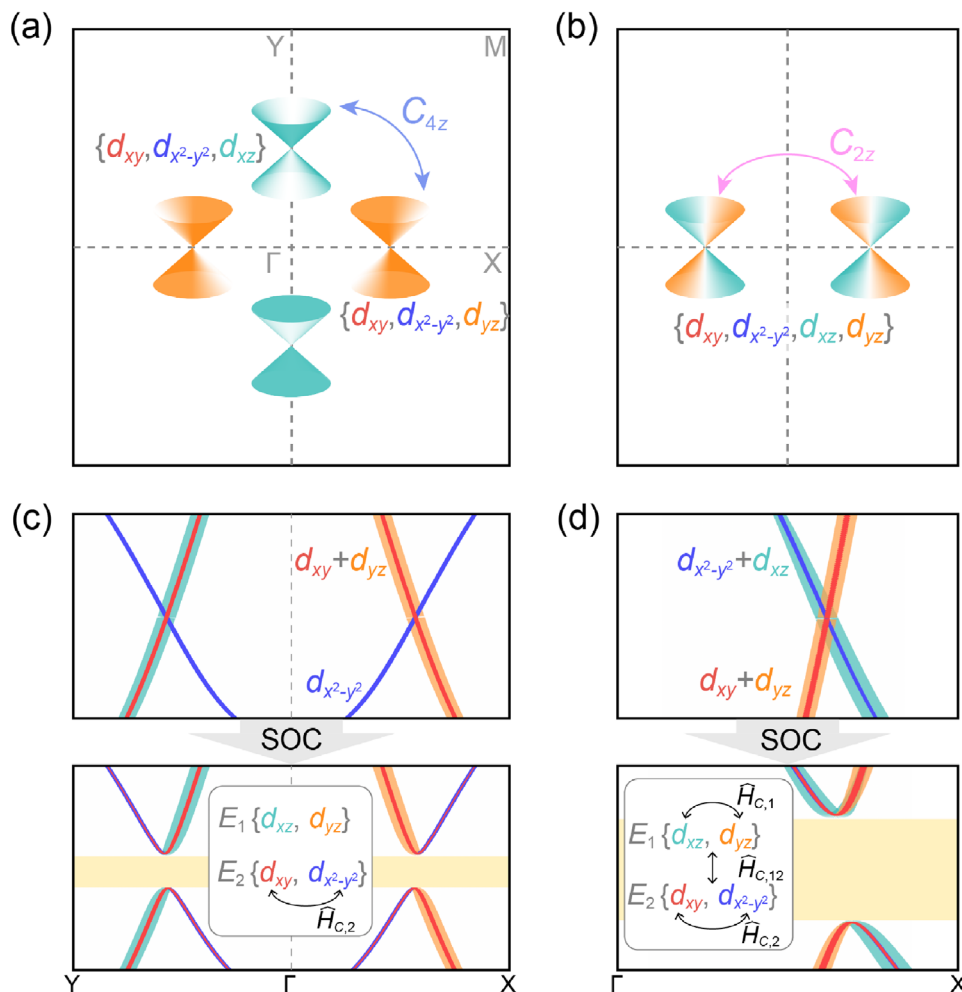


FIGURE 1 | Symmetry-lowering activated orbital hybridization and enhanced SOC gap. (a, b) Schematics of the Dirac-cone distributions in momentum space for the (a) $P4/nmm$ and (b) $Pmmn$ lattices. The C_{4z} symmetry in $P4/nmm$ separates the d_{xz} and d_{yz} components into symmetry-related cones, whereas in $Pmmn$ they coexist near the same cone after symmetry lowering. (c, d) Orbital-resolved Dirac dispersions along Γ -X and Γ -Y lines before and after including SOC. In $P4/nmm$, the orbitals in the E_1 set are separated at an individual cone, where only intra-set correlation $\hat{H}_{C,2}$ enhances the SOC gap, leading to a relatively small gap. In $Pmmn$, symmetry lowering activates hybridization between E_1 and E_2 sets within the same cone, where both intra- and inter-set correlation contribute to the SOC gap enhancement, resulting in a much larger SOC gap. Insets summarize the active correlation channels in the two cases. Here, $\hat{H}_{C,1}$ and $\hat{H}_{C,2}$ denote the intra-set correlation within E_1 and E_2 , respectively, while $\hat{H}_{C,12}$ denotes the inter-set correlation between them.

$C = 2$. Self-consistent calculations show that E_g is increased by correlation-enhanced effective SOC, as presented by black circles in Figure 2d, which agrees with previous results for a single E_2 set.

When C_{4z} is broken, only two Dirac cones remain in the Γ -X line (Figure 2b). Symmetry lowering prevents the separation of d_{xz} and d_{yz} , enabling the hybridization between the E_1 and E_2 sets in both cones (Note S2.2 and Figure S2). Introducing the SOC results in a QAH with a $C = -1$. As expected, the bandgap is substantially enhanced, as shown by blue squares in Figure 2d. Unlike the case of a single orbital set, the orbital polarization is almost quenched by large hopping here, suppressing the correlation-enhanced effective SOC. This implies that the orbital hybridization through symmetry-lowering can open a large E_g in the $Pmmn$ lattice per se in correlated systems (Note S2.3 and Figure S3).

2.3 | Materials Design

We explore realistic materials to verify our strategy, focusing on tetragonal 2D materials with a formula of MX (M = transition metals from the fourth and fifth periods, X = elements with atomic number < 36) (Figure 3a). High-throughput computations are performed following the same workflow as adopted in ref. [46], based on first-principles calculations [47, 48]; the calculation details are provided in Note S3. 141 thermally and dynamically stable MX monolayers are identified from a total of 171 compounds, including 84 nonmagnetic, 19 ferromagnetic, and 38 antiferromagnetic monolayers (Note S3 and Figures S4 and S5). Here, we seek QAHs from ferromagnets, although they recently have been reported in antiferromagnets [37, 49, 50] and ferrimagnets [51, 52].

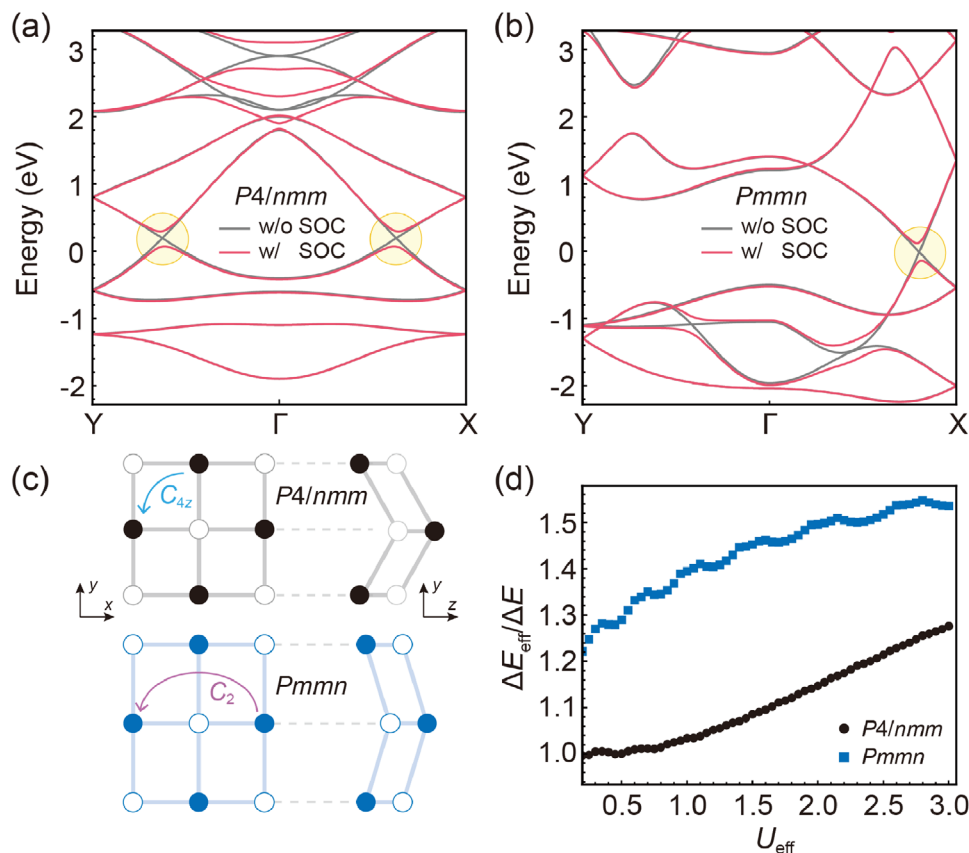


FIGURE 2 | Ten-band TB model for QAHIs. (a,b) Band structures of $P4/nmm$ (a) and $Pmmn$ (b) lattice models, with (red) and without (grey) considering SOC, respectively. (c) Schematic of TB model with $P4/nmm$ (upper panel) and $Pmmn$ (lower panel) symmetry. (d) Enhancement of the bandgap $\Delta E_{\text{eff}}/\Delta E$ as a function of correlation strength U_{eff} in the $P4/nmm$ (black circles) and $Pmmn$ (blue squares) lattices.

From the 19 ferromagnets, we screen out four MX ($M = \text{V}, \text{Ti}$, and $X = \text{P}, \text{As}$) as new QAHIs, where lattice constants, T_c , and E_g are collected in Table 1 (see also Note S3.1 and Table S1). The experimental feasibility of these tetragonal monolayers is supported by the successful synthesis of structural analogs. Specifically, while the tetragonal symmetry is the ground-state phase for iron-chalcogenides such as FeSe [53, 54] and FeTe [55], their realization in the monolayer limit relies on epitaxial growth on oxide substrates (e.g., SrTiO_3) to provide a 2D template. Moreover, this structural motif can also be experimentally stabilized in systems where the tetragonal phase is a metastable polymorph, as demonstrated by the recently reported layered tetragonal CoSe nanoplates and monolayers grown on $\text{SrTiO}_3(001)$ surfaces [56, 57]. These precedents prove that the tetragonal motif can be robustly maintained at the monolayer limit via substrate-mediated kinetic stabilization. Furthermore, these lattice constants (4.5–4.8 Å) exhibit high compatibility with several standard substrates, ensuring epitaxial growth feasibility. For instance, CdO [58, 59] provides a nearly perfect match for TiAs with a minimal mismatch of $\sim 1.3\%$ (Note S3.1 and Figure S6). Additionally, halide perovskites [60] with tunable lattice parameters could serve as effective buffer layers for these MX systems. Notably, beyond providing a growth template, the use of such substrates could also introduce epitaxial strain or interfacial interactions to further facilitate the symmetry-lowering transition from $P4/nmm$ to $Pmmn$, directly supporting the symmetry strategy proposed in this work.

Among these candidates, the 2D VAs exhibit the best combination of a T_c up to 850 K and an E_g as high as 1018 meV. The high T_c in 2D VAs can be understood by analyzing the orbital occupation. V atoms dominate the total magnetization and adopt an oxidation state of +3 with a valence electron configuration $3d^2 4s^0$. In a tetrahedral crystal field, the V 3d orbitals split into higher-energy triplet t_{2g} (d_{xz} , d_{yz} , d_{xy}) and lower-energy doublet e_g ($d_{x^2-y^2}$, d_{z^2}) (Note S3.1 and Figure S7b). According to Hund's rule, such a splitting results in a net magnetic moment of $2 \mu_B$ per V atom, consistent with our calculations. The substantial ferromagnetism arises predominantly from a d – p superexchange, which overwhelms a direct d – d hopping due to the near 90° V–As–V bond angle, marked as α in Figure 3a. This mechanism is further corroborated by real-space electronic and magnetic analyses (Note S3.1 and Figure S8), which reveal prominent charge accumulation and induced spin polarization along the V–As–V pathways, confirming the mediator role of As atoms. Then, simulated temperature-dependent magnetization using the Heisenberg model estimated a T_c of 850 K for the 2D VAs (Note S3.1 and Figure S9a).

Band structure of the VAs exhibits a Dirac cone at the Fermi energy in the Γ –X line in the spin-up channel, while an indirect bandgap of 2.35 eV exists in the spin-down channel (Figure 3b). The cone results from an accidental band crossing, rather than symmetry-enforced degeneracy. Although the SOC effect in 3d metals is generally weak, including the SOC gaps out of the

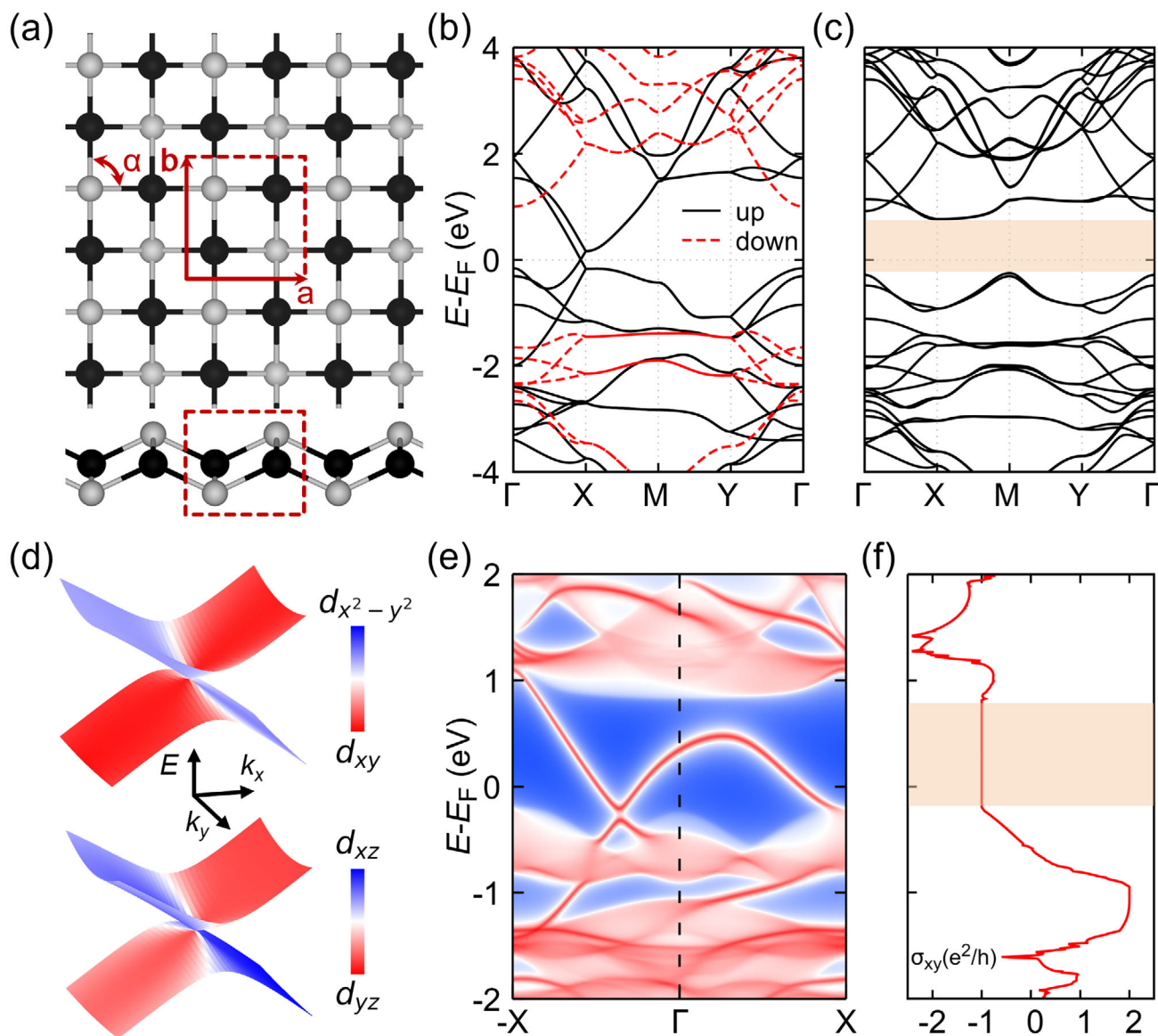


FIGURE 3 | Illustration of our strategy with the VAs monolayer. (a) Atomic structure of 2D VAs from top and side views. (b,c) Band structure of VAs without and with SOC. (d) Three-dimensional band structures projected onto V 3d orbitals without SOC, $k_x \in (0.41, 0.68)$, $k_y \in (-0.14, 0.14)$. (e) Topological edge calculated along the x-axis; and (f) anomalous Hall conductance σ_{xy} as a function of Fermi energy. The shaded regions in (c) and (f) denote the topological gap.

TABLE 1 | Calculated lattice constant, Curie temperature T_c , and bandgap E_g , with E_g specifically calculated by the HSE06.

Materials	a (Å)	b (Å)	T_c (K)	E_g (meV)
VAs	4.60	4.57	850	1018
VP	4.49	4.51	820	547
TiAs	4.76	4.76	360	144
TiP	4.66	4.66	290	85

cone by 1018 meV (Figure 3c) when using the HSE06 functional. Additional calculations with GGA+ U ($U = 4$ eV) show a bandgap

of 650 meV (Note S3.2 and Figure S12b), attesting the robustness of the results (Note S3.1 and Figure S10). The nontrivial topological nature is further verified in the calculated anomalous Hall conductance σ_{xy} , which shows a quantized plateau of e^2/h within the bulk gap, indicating $C = -1$ (Figure 3f). The calculated edge density of states presents a single chiral edge state within the bandgap (Figure 3e).

Since the 2D VAs has the lower $Pmmn$ symmetry, such a large bandgap opening is excellently coherent with our symmetry-lowering strategy as established by the above theory and TB model. To substantiate this point, we first analyze the orbital hybridization of V 3d orbitals near the Dirac point. Indeed, both the E_1 and E_2 sets are distributed near the Dirac cone and exhibit inter-set orbital hybridization (Figure 3d), which, as described

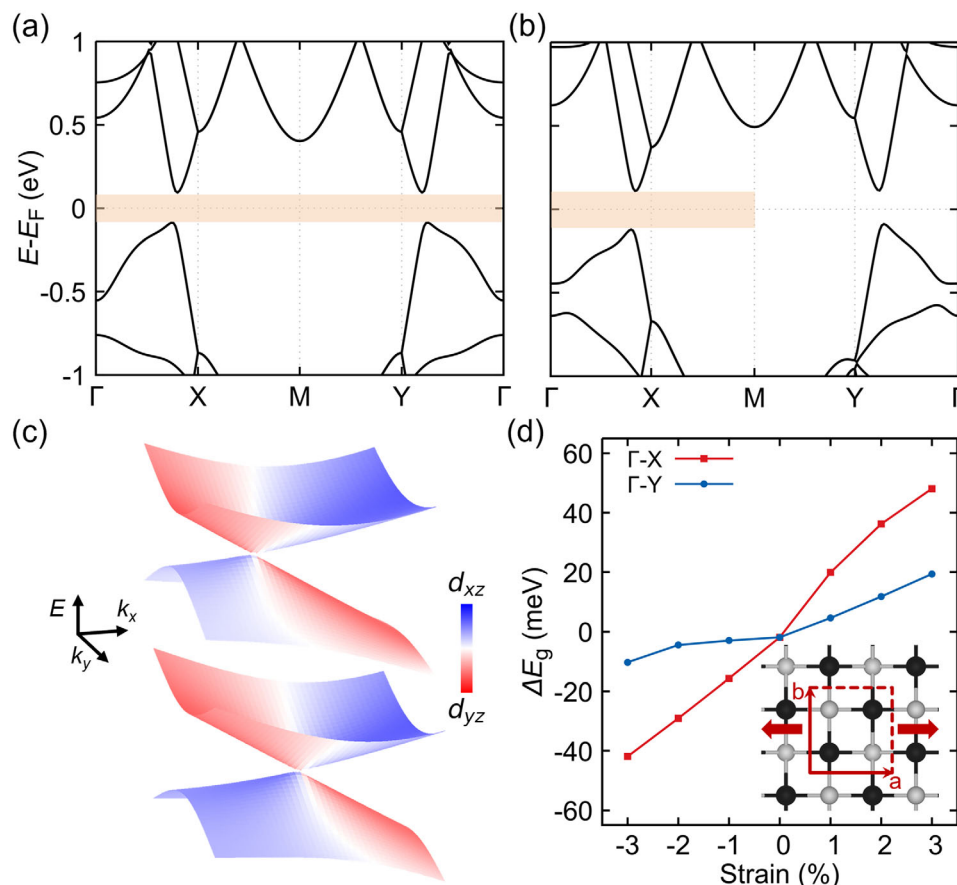


FIGURE 4 | Illustration of our strategy with the deformed TiTe monolayer. (a,b) Band structures of 2D TiTe with SOC under (a) zero and (b) 3% tensile strain, respectively. (c) Three-dimensional band structures projected onto Ti 3d orbitals without SOC (Upper panel: with zero strain, $k_x \in (0.46, 0.77)$, $k_y \in (-0.15, 0.15)$; Lower panel: with 3% strain, $k_x \in (0.45, 0.77)$, $k_y \in (-0.16, 0.16)$). (d) Changes in local bandgaps along the Γ -X and Γ -Y lines as functions of applied uniaxial strain to the 2D TiTe. The shaded regions in (a) and (b) denote the topological gap.

by our theory, enables a combined effect of the correlation and orbital hybridization to widen E_g . Next, similar analysis on an artificial 2D VAs with a $P4/nmm$ symmetry shows that only the E_1 set is distributed near the Dirac cone (Note S3.2 and Figure S17a–c), thereby involving no inter-set orbital hybridization. As a result, the opened bandgap in this artificial VAs monolayer is reduced to 857 meV with the HSE06 functional.

The bandgap increase by symmetry lowering is also reflected in another three newly predicted QAIs, namely the 2D VP, TiAs, and TiP (Note S3.2 and Figure S11). The former one has the $Pmmn$ symmetry and exhibits a nontrivial bandgap of up to 547 meV (Note S3.2 and Figure S13b), along with a T_c of 820 K; whereas the latter two adopt the higher $P4/nmm$ symmetry, thereby possessing E_g of merely 144 and 85 meV, respectively. For these two Ti-based 2D materials, their bandgaps of approximately 100 meV are attributed to the predominant contribution of E_2 set to the Dirac cone, with the E_1 set showing negligible involvement. This finding is consistently supported by both the d -orbital projection band structures (Note S3.2 and Figure S14g–i) and the three-dimensional band projection plots (Note S3.2 and Figure S16a,b). Our calculations with GGA+ U produce similar results.

Aside from the intrinsic enhancement effect in $Pmmn$ monolayers, our strategy can be utilized to widen E_g of $P4/nmm$ monolayers via symmetry manipulation by, for example, applied strain. To showcase this modulation, we take 2D TiTe as an example, which adopts the $P4/nmm$ symmetry. In this monolayer, a total of four Dirac cones are located along the Γ -X and Γ -Y axes (two on each axis) and are protected by mirror symmetries M_y and M_x , respectively. Including the SOC effect opens an E_g of 181 meV at the Dirac point (Figure 4a). An applied uniaxial tensile strain of 3% reduces the $P4/nmm$ to $Pmmn$ symmetry, resulting in an increase of E_g in the Γ -X line from 181 to 231 meV (Figure 4b). Orbital analysis suggests that the Dirac cone is mainly contributed by the E_2 set without strain, which provides channels for the first-order SOC. The applied strain enhances the contribution of E_1 set to the Dirac cone and, thus, strengthens the inter-set hybridization (Figure 4c), resulting in an increased E_g in the Γ -X line. However, a compressive strain narrows the local E_g (Figure 4d) despite the symmetry reduction, mainly because it leads to less contributed d orbitals to the Dirac cones (Note S3.2 and Figure S15).

The local bandgap in the Γ -Y line exhibits a weaker response to strain, attributed to the less contribution of $d_{x^2-y^2}$ and d_{yz} orbitals

to the Dirac cone in this particular direction. Consequently, the global band gap does not monotonically increase with applied strain. Even so, the effective modulation of local E_g by strain still provides solid proof for our symmetry-lowering strategy. The strain modulation is applicable to other 2D lattices, notably the hexagonal lattice. For example, a 3% uniaxial tensile strain can widen E_g by 50 meV for a hexagonal V_2O_3 monolayer, where only the E_1 set contributes to the Dirac cone but d_{xz} exhibits an increased weight by the tensile strain (Note S3.3 and Figures S18d and S19). Furthermore, this strategy is also validated in hexagonal FeS and FeSe monolayers, where symmetry-lowering results in a significant bandgap enhancement (Note S3.3 and Figures S20 and S21). Alternatively, the symmetry of a 2D material can be lowered by the substrate underneath as well, as exemplified by the V_2O_3 monolayer whose topological E_g can be increased by 70 meV by a $MgCl_2$ substrate [61].

3 | Conclusions

In conclusion, we have proposed a symmetry-lowering theory for designing large bandgap QAHIs with nonheavy elements. Our TB model verifies a significant enhancement on the bandgap of QAHIs by activating orbital hybridization near the Dirac cones through reducing the lattice symmetry, which can overwhelm previously reported SOC enhancement by orbital polarization. This theory is materialized in tetragonal VAs and VP monolayers from high-throughput screening, which are predicted to be high-temperature QAHIs. Moreover, our theory can guide the bandgap modulation in QAHIs by symmetry manipulation and potentially provide a new knob for controlling other SOC-related electronic states.

4 | Methods

4.1 | First-Principles Calculations

First-principles calculations were performed using the Vienna ab initio simulation package (VASP) [48] with the projector augmented wave (PAW) method [62]. The exchange-correlation energy is approximated by the generalized gradient approximation (GGA) in the Perdew–Burke–Ernzerhof (PBE) parametrization [63]. A plane-wave energy cutoff of 500 eV is used, and a vacuum space of 20 Å is introduced to prevent interactions due to the periodic boundary conditions. The energy and Hellmann–Feynman force convergence criteria are set to 10^{-5} eV and 10^{-2} eV/Å, respectively. For the monolayer unit cell and 2×2 supercell calculations, Γ -centered Monkhorst-Pack k-point meshes of $20 \times 20 \times 1$ and $10 \times 10 \times 1$ are used. The magnetocrystalline anisotropy energy (MAE) is evaluated with spin-orbit coupling (SOC) at a denser mesh of $48 \times 48 \times 1$. To account for the correlation effects of V and Ti 3d electrons, both the GGA + U approach [64] and HSE06 method [65] are employed. In the GGA + U method, the effective U value ($U_{\text{eff}} = U - J$) is used to describe the exchange-correlation strength of the 3d states. For consistency, U_{eff} values of 3 and 4 eV are used for Ti and V, respectively, unless stated otherwise. Specifically, these U_{eff} values were determined by benchmarking the GGA + U band structures against HSE06 results to accurately describe the electronic states near the Fermi level. Furthermore, the

influences of on-site interactions were checked by varying U_{eff} from 2.0 to 5.0 eV, which confirmed that the non-trivial band topology and bandgap enhancement are robust across a wide range of U_{eff} values. The Heyd–Scuseria–Ernzerhof (HSE) hybrid functional (HSE06 method) is specifically used for determining the band gap [65], which is most sensitive to correlation effects, ensuring more accurate results. Edge states and anomalous Hall conductance calculations are performed using maximally localized Wannier functions (MLWF) via Wannier90 [66, 67] and WannierTools [68] packages. Crystal structures are plotted by VESTA [69].

Author Contributions

Wanlin Guo: writing – review and editing. **Zhuhua Zhang:** funding acquisition, supervision, writing – original draft, writing – review and editing, conceptualization, resources, project administration, investigation. **Han Cai:** writing – original draft, investigation, conceptualization, data curation, writing – review and editing. **Qiushi Yao:** writing – review and editing, investigation, data curation. **Xiaoyu Xuan:** writing – review and editing, investigation, data curation. **Jiayu Li:** data curation, supervision, writing – original draft, writing – review and editing, conceptualization, investigation.

Acknowledgements

The work was supported by the National Natural Science Foundation of China (Nos. 12225205, 12261160367, U2441272, 12532020 and 52402190), the National Key Research and Development Program of China (No. 2024YFA1409600), Fundamental and Interdisciplinary Disciplines Breakthrough Plan of the Ministry of Education of China (No. JYB2025XDXM205), Natural Science Foundation of Jiangsu Province (Nos. BK20253025, BK20243044, BK20253057 and BK20243065), the Fundamental Research Funds for the Central Universities (No. NG2024026 and NP2024110), and the Research Fund of State Key Laboratory of Mechanics and Control for Mechanical Structures (Nos. MCMS-I-0422K01 and MCAS-I-0525K01). This work was partially supported by the High Performance Computing Platform of Nanjing University of Aeronautics and Astronautics. J. L. thanks Xiangang Wan for helpful discussions.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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